

2.4 BANDWIDTH, SCAN TIME & SNR

SPEED, CLARITY & THE BALANCING ACT

Bandwidth, scan time, and SNR are closely linked. Change one, and the others are affected.
The technologist must balance image quality with exam time and patient comfort.



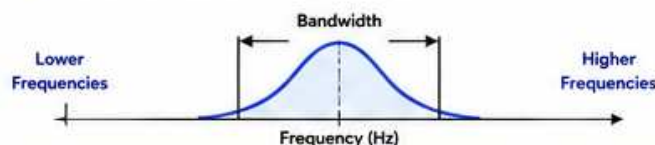
KEY CONCEPT

There is no free lunch in MRI.
Higher quality (higher SNR, higher resolution)
usually costs more time — or requires trade-offs.

- ↑ Increase / Improve
- ↓ Decrease / Reduce
- No / Minimal Effect

1 BANDWIDTH (BW) BASICS

Bandwidth is the range of frequencies the receiver accepts during signal acquisition.
It affects noise, chemical shift, and artifacts.



LOWER BANDWIDTH

- Less noise → higher SNR
- More chemical shift
- More susceptibility artifacts
- Longer scan time

Lower BW = Cleaner signal

HIGHER BANDWIDTH

- More noise → lower SNR
- Less chemical shift
- Less susceptibility artifacts
- Shorter scan time

Higher BW = Faster scan

2 HOW PARAMETERS AFFECT EACH OTHER

Parameter Change	Effect on SNR	Effect on Scan Time	Effect on Artifacts / Other
↓ Bandwidth (narrower)	↑ SNR (less noise)	↑ Scan time	↑ Chemical shift ↑ Susceptibility
↑ Bandwidth (wider)	↓ SNR (more noise)	↓ Scan time	↓ Chemical shift ↓ Susceptibility
↑ Matrix (more phase encode lines)	↑ SNR	↑ Scan time	↑ Resolution (smaller pixels)
↓ Matrix (fewer phase encode lines)	↓ SNR	↓ Scan time	↓ Resolution (larger pixels)
↑ NEX / Averages	↑↑ SNR (~√NEX)	↑ Scan time (linear)	— (minimal effect)
↓ NEX / Averages	↓ SNR	↓ Scan time	— (minimal effect)

3 SNR EXPLAINED

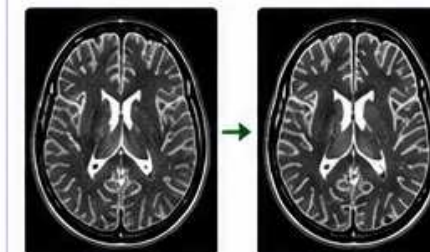
$$\text{SNR} = \frac{\text{Signal}}{\text{Noise}}$$

Higher SNR → better image quality (better detail, fewer artifacts)

SNR increases with:

- Lower bandwidth
- Higher matrix
- Higher NEX / averages
- Larger voxel size
- Stronger magnetic field (e.g., 3T > 1.5T)

SNR EXAMPLE (Same Anatomy)



4 BANDWIDTH EFFECT ON IMAGE QUALITY (T2-WEIGHTED AXIAL BRAIN)

LOW BANDWIDTH (±15.6 kHz)

- Higher SNR (less noise)
- More chemical shift
- More susceptibility
- Longer scan time



MEDIUM BANDWIDTH (±31.2 kHz)

- Balanced SNR
- Moderate chemical shift
- Moderate susceptibility
- Moderate scan time



HIGH BANDWIDTH (±62.5 kHz)

- Lower SNR (more noise)
- Less chemical shift
- Less susceptibility
- Shorter scan time



Cleaner Image (Higher SNR)

Faster Scan (Lower BW Penalty)

5 WHAT AFFECTS SCAN TIME?

$$\text{Scan Time} \propto \frac{\# \text{ of Phase Encodes} \times \text{NEX}}{\text{Bandwidth}} \times \text{Other Factors}$$

# of Phase Encodes	= Matrix (phase) + extra lines (e.g., partial Fourier)
NEX / Averages	= Number of excitations
Bandwidth	= Receiver bandwidth (Hz/pixel)
Other Factors	= # of slices, TR, TE, sequence type, acceleration

TO REDUCE SCAN TIME

- Increase bandwidth
- Decrease matrix (within limits)
- Decrease NEX (if SNR allows)
- Use faster sequences
- Use parallel imaging (e.g., SENSE, GRAPPA)
- Use partial Fourier
- Reduce # of slices (if clinically acceptable)



6 VISUAL EXAMPLES OF TRADE-OFFS (SAME PATIENT, T2 AXIAL BRAIN)

	Baseline (Balanced)	↓ Bandwidth (Narrow)	↑ Bandwidth (Wide)	↓ Matrix (128 x 128)	↑ Matrix (512 x 512)	↓ NEX (1)	↑ NEX (4)
Image							
SNR	Good	Better	Lower	Lower	Better	Lower	Better
Detail / Resolution	Good	Good	Good	Poor	Excellent	Lower	Excellent
Scan Time	~ 5:00 min	~ 6:30 min	~ 2:30 min	~ 2:30 min	~ 8:00 min	~ 1:20 min	~ 9:30 min

7 PRACTICAL GUIDELINES



KNOW THE CLINICAL GOAL

Detail vs. speed? Diagnose vs. screening?
Choose parameters that support the goal.



START WITH PROTOCOL STANDARDS

Use proven protocols as a baseline.
Adjust one parameter at a time.



OPTIMIZE, DON'T MAXIMIZE

More is not always better.
Find the best balance for your patient.



CONSIDER PATIENT FACTORS

Motion, size, ability to hold still,
and clinical condition matter.

8 ENGINEER CHECKLIST

- ☒ Is the bandwidth appropriate for the sequence and anatomy?
- ☒ Is scan time reasonable for the clinical need?
- ☒ Is SNR adequate for diagnostic quality?
- ☒ Are artifacts (chemical shift, susceptibility) acceptable?
- ☒ Are parameter choices consistent with protocol and hardware?
- ☒ Document and communicate any changes made.



KEY TAKEAWAY

Bandwidth, scan time, and SNR are interconnected.
Excellent images come from smart parameter choices,
not just pushing numbers to the extremes.
Balance is the engineer's advantage.